Classical and ∞-categorical Adams representability

Brown representability: $\mathcal{C}$ ∞-cat $\rightsquigarrow$ $h\mathcal{C}$ homotopy category

$F : h\mathcal{C}^{op} \longrightarrow \mathrm{Set}$ is **weakly left exact** if:

(W1) F preserves products of $\mathcal{C}$: $\quad F(\coprod_\alpha X_\alpha) \xrightarrow{\cong} \prod_\alpha F(X_\alpha)$

(W2) For every pushout in $\mathcal{C}$
$$\begin{tikzcd} Z \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & Y \arrow[d] \\ X \arrow[r] & X \sqcup_Z Y \end{tikzcd}$$
the induced map $F(X \sqcup_Z Y) \longrightarrow F(X) \times_{F(Z)} F(Y)$ is surjective.

F is **representable** if $\exists B \in \mathcal{C}$ s.t. $[-, B] \cong F(-)$.

homotopy classes of maps

Under what conditions on $\mathcal{C}$,

$$F \text{ is weakly left exact} \iff F \text{ is representable} \; ?$$

$\Longleftarrow$ always

$\mathcal{C}$ = pointed connected spaces [Brown, 62] ✓

Ex. $F = H^n : h\mathcal{S}_*^{\geq 1\,op} \longrightarrow \mathrm{Set}$ generalized cohomology theory

$\mathcal{C}$ = spectra [Margolis, 83] ✓

Ex. $F = H^* : h\mathcal{S}p^{op} \longrightarrow \mathrm{Set}$ generalized cohomology theory

We say that $\mathcal{C}$ is **weakly generated** by a set $\{S_\alpha\}$ if

$f : X \to Y$ in $\mathcal{C}$ s.t. $\begin{matrix}[S_\alpha, X] \to [S_\alpha, Y] \\ \text{is an iso for all } \alpha\end{matrix} \overset{\text{always}}{\Longleftrightarrow} f$ is a weak equivalence

Brown representability theorem. [Jardine, 09] [Lurie, 17]
$\mathcal{C}$ presentable ∞-cat weakly generated by a set of compact objects.
Then F is weakly left exact $\Longleftrightarrow$ F is representable.

Not included: Given a gen. homology theory H_*, by SW-duality there is a gen. cohomology theory H^* only for finite spaces. Brown does not apply (not cocomplete and extensions of F are not weakly left exact)

$H^n : S_*^{fin, \geq 1} \longrightarrow Set$ is representable? Yes [Adams, 71]

Adams representability theorem. [Hayoun, 23]

C pointed finitely cocomplete ∞-cat with hC countable and $Ind(C)$ weakly gen. by a set of objects of C

$F : hC^{op} \to Grp$ s.t. $\hat{F} : hInd(C)^{op} \to Set$ extension of F

Then F is weakly left exact $\iff$ F is representable in Ind(C)

$$\Updownarrow$$

$$\exists B \in Ind(C), [-, B] \cong F(-)$$

Also, $[-, B] \to \hat{F}(-)$ is objectwise surjective and B is unique up to iso.

Lemma 1. All the previous assumptions + F is weakly left exact. For every $A \in \mathrm{Ind}(C)$ and $a \in \hat{F}(A)$, there exists a map $f : A \to B$ in $\mathrm{Ind}(C)$ and $b \in \hat{F}(B)$ lifting a and inducing bijections

$$[S_\alpha, B] \to F(S_\alpha) \text{ for all } \alpha \in A.$$

Lemma 2. C pointed finitely cocomplete ∞-cat with hC countable. $F : hC^{op} \to \mathrm{Grp}$ weakly left exact and $\hat{F} : h\mathrm{Ind}(C)^{op} \to \mathrm{Set}$ extension of F. For every pushout

$$\begin{tikzcd} Z \arrow[r] \arrow[d] & Y \arrow[d] \\ X \arrow[r] & X \sqcup_Z Y \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$$

in $\mathrm{Ind}(C)$ where $Z \cong \coprod_i A_i$ and $A_i \in \mathcal{G}$, the induced map $\hat{F}(X \sqcup_Z Y) \to \hat{F}(X) \times_{\hat{F}(Z)} \hat{F}(Y)$ is surjective.

Proof theorem. Apply Lemma 1 to $A=\emptyset$, then

$\exists B \in \mathrm{Ind}(C)$ and $b \in \hat{F}(B)$ s.t. $[S_\alpha, B] \to F(S_\alpha)$ bijections.

$\Updownarrow$

We want $[-,B] \xrightarrow{b} \hat{F}(-)$ to be surjective and injective on $\mathcal{C}$

Surjectivity: $X \in C$ and $x \in \hat{F}(X)$. By Lemma 1 applied to $(b,x) \in \hat{F}(B \sqcup X)$,

$\Rightarrow \exists\ B \sqcup X \to X'$ and $x' \in \hat{F}(X')$ s.t. ...

$$\Rightarrow B \xrightarrow{\varphi} X' \Rightarrow [S_\alpha, B] \xrightarrow{\varphi^*} [S_\alpha, X'] \Rightarrow \varphi^* \text{ are bijections for all } \alpha$$

$$\begin{aligned} F(B) &\longrightarrow F(X') \\ b &\longmapsto x' \end{aligned}$$

$$\begin{tikzcd} {[S_\alpha, B]} \arrow[rr, "\varphi^*"] \arrow[dr, "\cong"] & & {[S_\alpha, X']} \arrow[dl, "\cong"'] \\ & F(S_\alpha) & \end{tikzcd}$$

$\Rightarrow \varphi$ is an iso in $\mathrm{Ind}(C)$

Then $X \to X' \cong B$ is the preimage of x $\square$

Injectivity. $X \in C$ and $f, g : X \to B$ two preimages of $x \in F(X)$ $f^*(b) = x = g^*(b)$. There is a pushout in $\mathrm{Ind}(C)$

$$\begin{tikzcd} X \sqcup X \arrow[r, "f+g"] \arrow[d, "\nabla"'] & B \arrow[d] \\ X \arrow[r] & W \arrow[ul, phantom, "\ulcorner"] \end{tikzcd}$$

By Lemma 2, $\exists\, w \in \hat{F}(W)$ lifting x and b.

By Lemma 1, $\exists\, W \to W'$ and $w' \in \hat{F}(W')$ lifting w and inducing bijections $[S_\alpha, W'] \to F(S_\alpha)$

The composite $h : B \to W \to W'$ induces bijections on $[S_\alpha, -]$ for all α then h is an iso in $\mathrm{Ind}(C)$. Since $h \circ f = h \circ g$, $f = g$. $\square$

Proof Lemma 1. We want a sequence $A \to B_0 \to B_1 \to \cdots$ in $\mathrm{Ind}(C)$ and compatible elements $b_n \in \hat{F}(B_n)$ lifting a.

- $B_0 = A \sqcup \coprod_{\alpha, s \in F(S_\alpha)} S_\alpha$. Because $\hat{F}$ sends coproducts to products

  $\exists\, b_0$ as the tuple of a and all $s \in F(S_\alpha)$ for all α and then

  $[S_\alpha, B_0] \to F(S_\alpha)$ is surjective for all α.

- Assume (B_n, b_n). R_α be equivalence relation on $[S_\alpha, B_n]$ s.t.

  b_n induces injective maps $[S_\alpha, B_n]/R_\alpha \longrightarrow F(S_\alpha)$.

Define B_{n+1} as the pushout in $\mathrm{Ind}(C)$

$$\begin{tikzcd}
\coprod_{\alpha, \pi \in R_\alpha} (S_\alpha \sqcup S_\alpha) \arrow[r, "\pi"] \arrow[d, "\nabla"] & B_n \arrow[d] \\
\coprod_{\alpha, \pi \in R_\alpha} S_\alpha \arrow[r] \arrow[ur, phantom, "\ulcorner"] & B_{n+1}
\end{tikzcd}$$

By Lemma 2, $\exists\, b_{n+1} \in \hat{F}(B_{n+1})$ lifting b_n

$B = \mathrm{colim}\, B_n$ and $b \in \hat{F}(B)$ lifting b_n

$[S_\alpha, B] \longrightarrow F(S_\alpha)$ surjective because the composite $[S_\alpha, B_0] \to [S_\alpha, B] \to F(S_\alpha)$ is surjective

Injectivity! $f, g : S_\alpha \to B$ s.t. $f^*(b) = g^*(b)$. S_α compact $\Rightarrow$ f, g factor through $f_n, g_n : S_\alpha \to X_n$ s.t. $f_n^*(x_n) = g_n^*(x_n)$ $\Rightarrow$ $f = g$ $\square$

Proof Lemma 2. Using Zorn's lemma, we can reduce the statement to $Z \in C$. Fix $(x,y) \in \hat{F}(X) \times_{\hat{F}(Z)} \hat{F}(Y)$ and want $w \in \hat{F}(W)$ lifting x and y, where $W = X \sqcup_Z Y$.

Define $\mathrm{Lift}_{x,y} : C^{op}_{Z//X} \times C^{op}_{Z//Y} \longrightarrow \mathrm{Set}$ sending factorizations $Z \to X' \to X$ and $Z \to Y' \to Y$ with $X', Y' \in C$ to the set of elements in $F(X' \sqcup_Z Y')$ lifting x and y.

W is the filtered colimit of the diagram $C_{Z//X} \times C_{Z//Y} \to C$ sending (X', Y') to $X' \sqcup_Z Y'$.

$\Rightarrow$ $w \in \hat{F}(W)$ is an element in the limit of $\mathrm{Lift}_{x,y}$

$\Rightarrow$ We want to show $\lim \mathrm{Lift}_{x,y}$ non-empty

It suffices to find a factorization of $\mathrm{Lift}_{x,y}$ by $A^{op} \to \mathrm{Set}$ s.t. limit of $A^{op} \to \mathrm{Set}$ is non-empty and A is a filtered poset.

$\mathrm{Lift}_{x,y}$ identifies parallel morphisms $\Rightarrow$ factors through homotopy 0-category

A poset s.t. objects same as $C_{Z//X} \times C_{Z//Y}$

and morphisms $(X', Y') \leq (X'', Y'')$ if $\forall\, X' \to X''' \leftarrow X''$ $\;Y' \to Y''' \leftarrow Y''$

the surj. $\mathrm{Lift}_{x,y}(X''', Y''') \to \mathrm{Lift}_{x,y}(X', Y')$ factors through

$$\mathrm{Lift}_{x,y}(X''', Y''') \to \mathrm{Lift}_{x,y}(X'', Y'')$$

Then $\mathrm{Lift}_{x,y}$ factors as

essent. surjective

$$C^{op}_{Z//X} \times C^{op}_{Z//Y} \to h_0 C^{op}_{Z//X} \times h_0 C^{op}_{Z//Y} \to A^{op} \xrightarrow{p} \mathrm{Set}$$

p sends (Z, Z) to a point and any morphism to a surjection

Lemma. A countable filtered poset $+ F : A^{op} \to \mathrm{Set}$ non-empty sending all morphisms to surjections, then the limit of F is non-empty

By the previous lemma, it suffices with showing A is countable.

Use pointed + F group-valued: there is an exact sequence of groups

$$F(X' \sqcup Y') \leftarrow F(X' \sqcup_Z Y') \leftarrow F(\Sigma Z) \leftarrow F(\Sigma(X' \sqcup Y'))$$

- $\mathrm{Lift}_{x,y}(X', Y')$ is a coset of the image of $F(\Sigma Z)$ in $F(X' \sqcup_Z Y')$
- The image of $F(\Sigma Z)$ in $F(X' \sqcup_Z Y')$ is the quotient of $F(\Sigma Z)$ by the image of $F(\Sigma(X' \sqcup Y'))$

$\Rightarrow (X', Y') \leq (X'', Y'') \Leftrightarrow$ the image of $F(\Sigma(X' \sqcup Y'))$ in $F(\Sigma Z)$ contains the image of $F(\Sigma(X'' \sqcup Y''))$

Use hC countable: Since $C_{\Sigma Z/}$ has countably many iso classes

$\Rightarrow A$ has countably many iso classes. $\square$

Cor. $\mathcal{C}$ symm. monoidal pointed finitely cocomplete ∞-cat
$\mathrm{Ind}(\mathcal{C})$ weakly gen. by a set of objects of $\mathcal{C}$
$F : h\mathcal{C}^{op} \to \mathrm{Grp}$ weakly left exact and $\hat{F} : h\mathrm{Ind}(\mathcal{C})^{op} \to \mathrm{Set}$ extension of F
Then, $\exists E \in \mathrm{Ind}(\mathcal{C})$ and a nat. iso $[1, - \otimes E] \xrightarrow{\sim} \hat{F}$

$f, g : X \to Y$ in $\mathrm{Ind}(\mathcal{C})$ **weakly homotopic** if $\forall K \in \mathcal{C}$ and $h : K \to X$
$f \circ h \sim g \circ h$. Denote $[X, Y]_w$ the set of weak hom. classes
and $h_w \mathrm{Ind}(\mathcal{C})$ 1-cat with homsets $[X, Y]_w$ $(h\mathcal{C} \subseteq h_w \mathrm{Ind}(\mathcal{C}))$

Adams $\Rightarrow$ $\forall X \in \mathrm{Ind}(\mathcal{C})$, $[X, B] \to \hat{F}(X)$ induces an iso $[X, B]_w \cong \hat{F}(X)$

$$\Rightarrow \mathrm{Grp}(h_w \mathrm{Ind}(\mathcal{C})) \xrightarrow{\sim} \mathrm{Fun}_{wlex}(\mathcal{C}^{op}, \mathrm{Grp})$$

Brown $\Rightarrow$ $h\mathrm{Ind}(\mathcal{C}) \xrightarrow{\sim} \mathrm{Fun}^{\pi}_{wlex}(\mathrm{Ind}(\mathcal{C})^{op}, \mathrm{Set})$

If C is additive: $h_w \mathrm{Ind}(C) \xrightarrow{\sim} \mathrm{Fun}_{wlex}(C^{op}, \mathrm{Set})$

C pointed finitely cocomplete ∞-cat $\quad SW(C) = \mathrm{colim}(C \xrightarrow{\Sigma} C \xrightarrow{\Sigma} \dots)$

$$\mathrm{Ind}(SW(C)) = \mathrm{Sp}(\mathrm{Ind}(C))$$

Cohomology theory on C = weakly left exact functor $SW(C)^{op} \to \mathrm{Set}$
(wlex functor $H^n : C^{op} \to \mathrm{Set}$ with isom $H^n \cong H^{n+1} \circ \Sigma$)

Homology " " " = " " " " $SW(C) \to \mathrm{Set}$

$$h_w \mathrm{Sp}(\mathrm{Ind}(C)) \xrightarrow{\sim} \mathrm{CohTh}(C)$$

$$h_w \mathrm{Sp}(\mathrm{Ind}(C)) \xrightarrow{\sim} \mathrm{HomTh}(C)$$

$\mathcal{C}$ category with finite coproducts

$S \in \mathcal{C}$ **admits a cogroup structure** if there is a multiplication map $S \to S \sqcup S$ s.t. $\forall Y \in \mathcal{C}$,

$$\mathrm{Hom}(S, Y) \times \mathrm{Hom}(S, Y) \cong \mathrm{Hom}(S \sqcup S, Y) \longrightarrow \mathrm{Hom}(S, Y)$$

endows $\mathrm{Hom}(S, Y)$ with a group structure

An ∞-cat $\mathcal{C}$ is generated by h-cogroups under finite colimits if it admits all finite colimits and contains an small collection of objects $\{S_\alpha\}$ s.t. (1) Each S_α admits a cogroup structure in $h\mathcal{C}$

(2) $\mathcal{C}$ is generated by $\{S_\alpha\}$ under finite colims and retracts

$S_*^{fin, \geq 1}$ generated by S^1. Pres. Stable cat every object has cogroup.

$\Rightarrow$ Weakly generated by compact objects